PDF Engine Feature Lab

This PDF is a practical test corpus in one document. It intentionally combines many frequently problematic PDF features: font encodings, vector painting operators, images and masks, transparency, annotations, AcroForms, page boxes, bookmarks, embedded files, and custom parser objects. It is not a complete substitute for ISO 32000 conformance testing.

Page	Feature group	Representative targets
1	Manifest and navigation	document info metadata, outline/bookmarks, internal named destinations, page lab
2	Text and fonts	Base 14 Type 1 fonts, embedded TrueType subsets, CID CJK font, Unicode text...
3	Vector graphics and path painting	lines, curves, arcs, miter/round/bevel joins...
4	Color, gradients, transparency, blend modes	DeviceGray, DeviceRGB, DeviceCMYK, linear/radial shadings...
5	Images and XObjects	JPEG/DCT image, PNG/Flate image, alpha/soft mask image, 1-bit image...
6	Page geometry	MediaBox, CropBox, BleedBox, TrimBox...
7	Annotations and links	URI link annotation, internal GoTo link, text annotation, free-text annotation..
8	AcroForm fields	text field, checkbox, radio group, combo box...
9	Tables, clipping, overprint-style stress	table layout, dense strokes, tiny text, rotated text...
10	Embedded data and parser objects	embedded file attachment, custom dictionary, booleans, integers...

Expected parser-only objects

Page 10 also contains an embedded JSON manifest and a custom /PDFSyntaxDataCharacteristics dictionary containing primitive PDF object types plus a custom stream object.

Text and fonts

Base 14 fonts, embedded TrueType, CID font, Unicode, text-state controls, and text render modes.

Helvetica: The quick brown fox jumps over 13 lazy dogs. 0123456789

Helvetica-Bold: The quick brown fox jumps over 13 lazy dogs. 0123456789

Helvetica-Oblique: The quick brown fox jumps over 13 lazy dogs. 0123456789

Times-Roman: The quick brown fox jumps over 13 lazy dogs. 0123456789

Times-BoldItalic: The quick brown fox jumps over 13 lazy dogs. 0123456789

Courier: The quick brown fox jumps over 13 lazy dogs. 0123456789

Courier-BoldOblique: The quick brown fox jumps over 13 lazy dogs. 0123456789

[redacted]: [redacted] [redacted] 1234 ([redacted] [redacted])

[redacted] [redacted] [redacted] [redacted] [redacted]

Embedded Unicode fonts and CID font

DejaVuSans embedded TrueType: Latin, accents, Greek, math: café naïve Ελληνικά ∑ ∫ √ ≤ ≥

NotoSans embedded TrueType: arrows and symbols: •

CID CJK font sample: 日本語のテキスト 漢字 かな カナ PDF 表示テスト

: العربية اختبار

Text object state: char spacing, word spacing, horizontal scaling, rise

Normal text line

C h a r s p a c i n g = 3

Word spacing visibly wider

Horizontal scale = 70 percent

Horizontal scale = 130 percent

Raised baseline

and lowered continuation

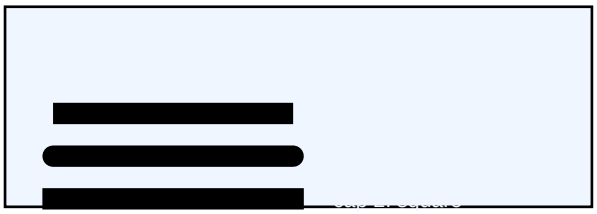
Text render modes

Ag fill Ag stroke Ag fill+stroke Ag Ag Ag

Note: text render modes 4-6 set text clipping; implementations vary in how clearly they expose the clip effect visually.

Vector graphics and path painting

Paths, curves, joins, caps, dash arrays, even-odd fill, clipping, gradients, transformations.

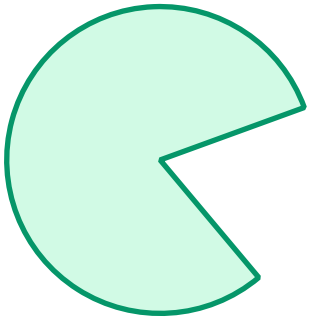




Dash pattern: 8 on / 4 off



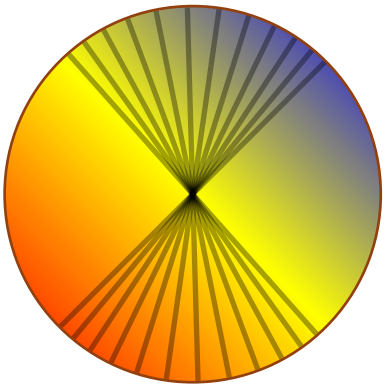
Cubic Bezier curves



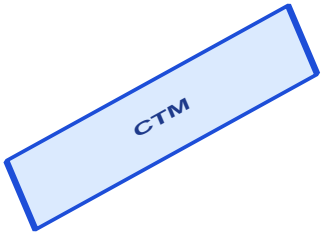
Arc/wedge path



Filled star path



Circular clip + linear shading



Transformed rectangle: translate + rotate + scale + skew

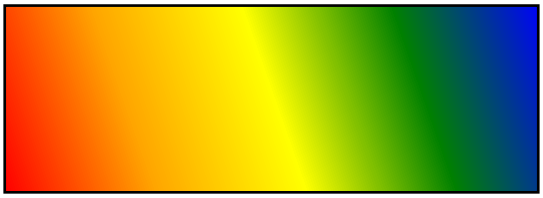
Color, gradients, transparency, and blend modes

Device color spaces, shadings, alpha state, and common blend modes.

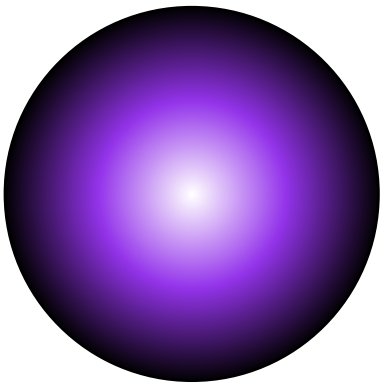


DeviceGray 25% | DeviceRGB red | DeviceRGB green | DeviceRGB blue | DeviceCMYK

Shadings

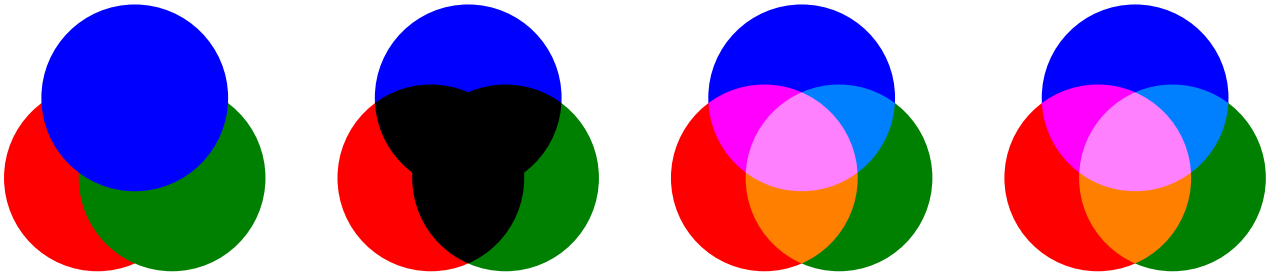


Linear gradient / axial shading



Radial shading

Alpha constants and blend modes



Normal | Multiply | Screen | Difference

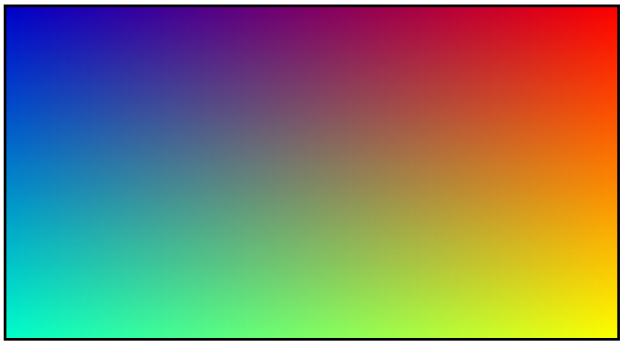


Stroke alpha = 0.35 overlapping diagonal strokes

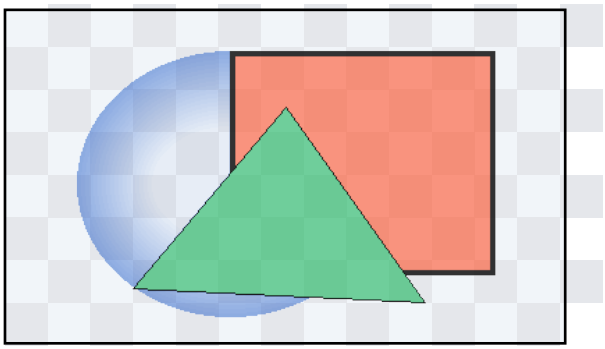
Images and XObjects

DCT/JPEG, Flate/PNG, alpha masks, monochrome images, scaling, rotation, clipping.

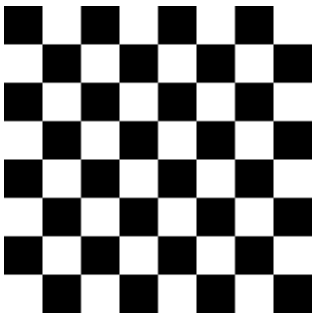
JPEG / DCTDecode, progressive source



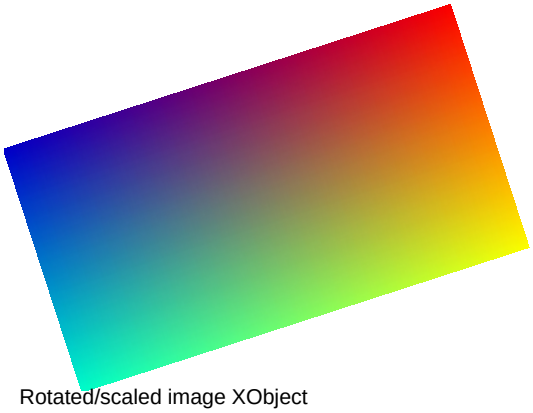
RGBA PNG with transparency / soft mask



1-bit image and transformations



1-bit checker image



Rotated/scaled image XObject

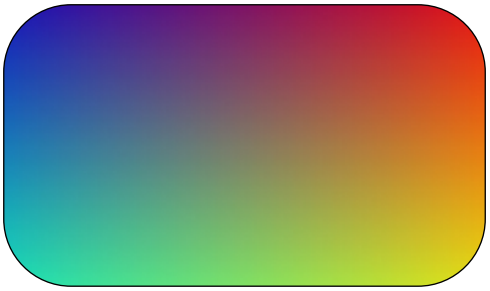


Image clipped to rounded rectangle


BleedBox
CropBox
Page geometry: boxes, rotation, non-default size
This page is landscape Letter with /Rotate 90 and explicit Crop/Bleed/Trim/Art boxes.
TrimBox
ArtBox
Rotated Page Geometry Test
Check /MediaBox, /CropBox, /BleedBox, /TrimBox, /ArtBox, and /Rotate handling.
Some PDF APIs expose box coordinates before page rotation; renderers apply rotation in display spa

Annotations and links

URI link, internal GoTo link, note, free text, highlight, ink annotation.

External URI link rectangle: https://example.com

Open example.com

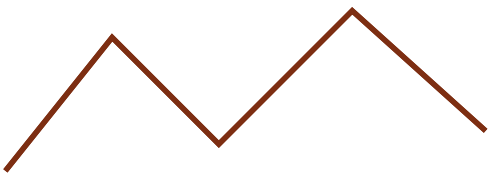
Text annotation icon

Internal link to page 1 manifest

FreeText annotation with valid /DA string.

FreeText annotation box above; /FreeText annotation added in post-processing

This sentence has a highlight annotation spanning part of the text.



Ink annotation path above; /Ink annotation added in post-processing

AcroForm fields

Interactive form fields with widget annotations and appearance streams.

Text field: Editable text value

Checkboxes: ☑ checked ☐ unchecked

Radio group: ◉ A ○ B ○ C

Combo box: Beta

List box:

One
Two
Three

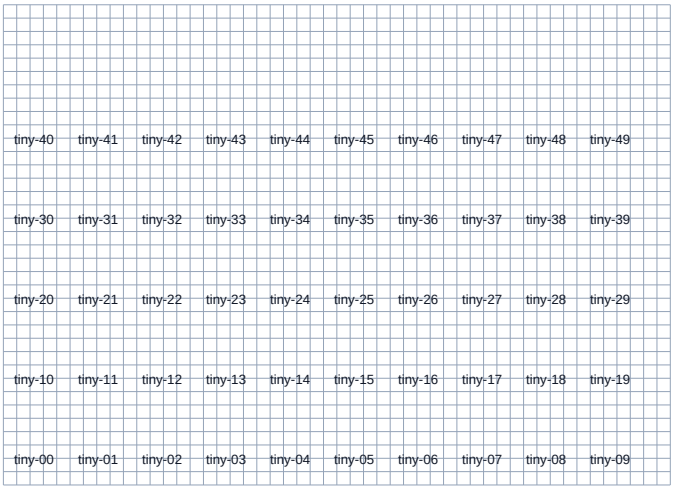
Expected checks

- AcroForm dictionary exists in catalog
- Each field has a /Widget annotation
- Default appearances render without needing user interaction
- Text extraction should not confuse field values with static labels

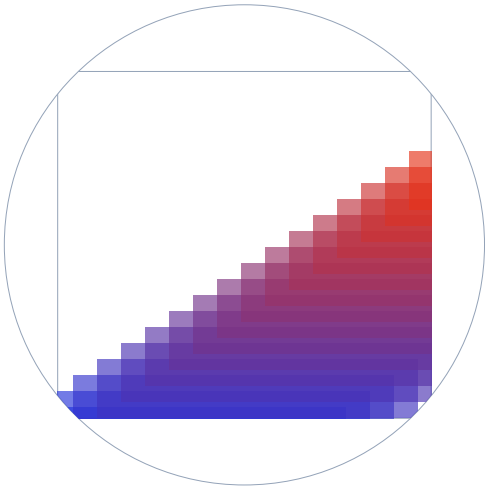
Tables, clipping, tiny text, and stress rendering

Dense strokes, nested clipping, rotated text, and table layout.

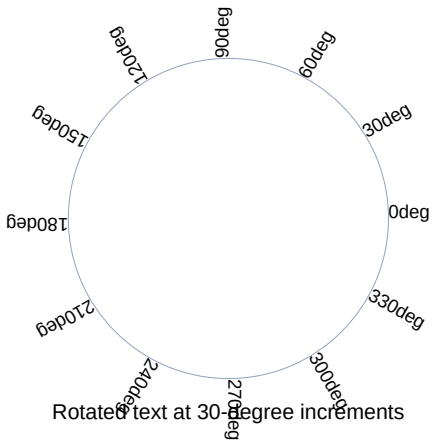
Operator	Visual target	Notes
BT/ET	text objects	font + text matrix
m/l/c/h	paths	move/line/curve/close
S/f/B	painting	stroke/fill/fill+stroke
q/Q/cm	graphics state	save/restore/matrix
Do	XObject image	draw image XObject
W/n	clipping	clip then end path



0.2pt dense grid + 5pt text



Nested clipping: circle ∩ rectangle



Rotated text at 30-degree increments

Embedded data and parser object types

This page is visually simple; post-processing injects custom primitive objects and an embedded JSON file.

Parser-focused structures

- The document catalog contains an embedded file attachment: pdf_engine_feature_lab_manifest.json.
- Page 1 contains a custom /PDFSyntaxDataCharacteristics dictionary.
- That dictionary contains booleans, integers, real numbers, names, literal strings, byte/hex strings, arrays, nested dictionaries, null, and an indirect custom stream.
- This page gives your engine a visible marker while the low-level PDF object tree gives your parser a non-visual target.

```
/PDFSyntaxDataCharacteristics <<
  /BooleanTrue true
  /Integer 123456
  /Real -45.6789
  /LiteralString (Literal string with parentheses and newline)
  /ByteString <4865782d7374796c65206279746573>
  /Array [1 2.5 /Name (text) << /Nested true >> null]
  /CustomStream 42 0 R
>>
```

11. Comments / Sticky Note Annotation Stress Page

This page contains 96 /Subtype /Text annotations, treated by viewers as sticky-note comments. Each visible marker C001-C096 has a real annotation object with author, subject, dates, color, icon, flags, popup, and unique /NM metadata. An additional uncompressed no-op content stream with 96 raw % PDF comment lines is appended for syntax-parser testing.

Expected parser checks: page_count=11; page 11 annotation_count=96; page 11 raw comment tokens=96 (RAW-PDF-COMMENT-001..096).



Note: PDF syntax comments beginning with % do not render. They are inserted into a separate uncompressed content stream after this visible page content.